PROJECT REPORT
PHISHING AWARENESS SIMULATION PLATFORM

Course: Cybersecurity & Ethical Hacking

System Architecture: Laravel (PHP) / MySQL / Mailtrap

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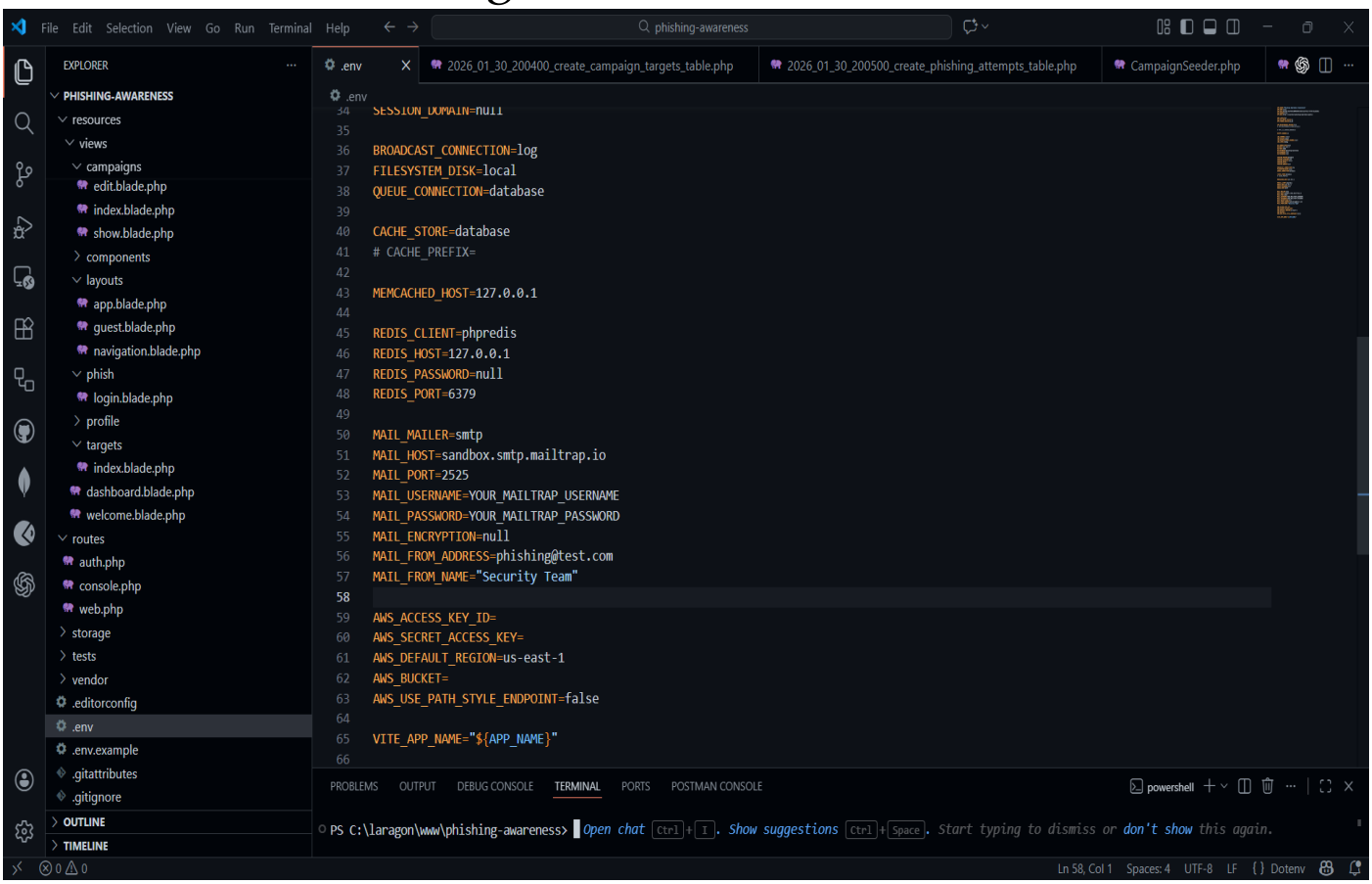
1. EXECUTIVE SUMMARY

This project documents the development and successful deployment of a **Phishing Awareness Simulation** platform. Built using the **Laravel** framework, the system provides a controlled environment to simulate social engineering attacks. By integrating **Mailtrap** for SMTP sandboxing, the platform allows for the safe delivery of phishing emails, tracking of user interactions (clicks), and the secure logging of simulated credential harvesting—all designed for organizational security training and awareness.

2. SYSTEM ARCHITECTURE & TOOLS

To ensure a professional and secure simulation, the following stack was utilized:

- **Backend:** Laravel Framework (MVC Architecture).
- **Database:** MySQL (managed via phpMyAdmin) for attempt logging and campaign management.
- **Email Sandbox:** Mailtrap (to prevent accidental delivery to real-world inboxes).
- **Environment:** Laragon Local Server.

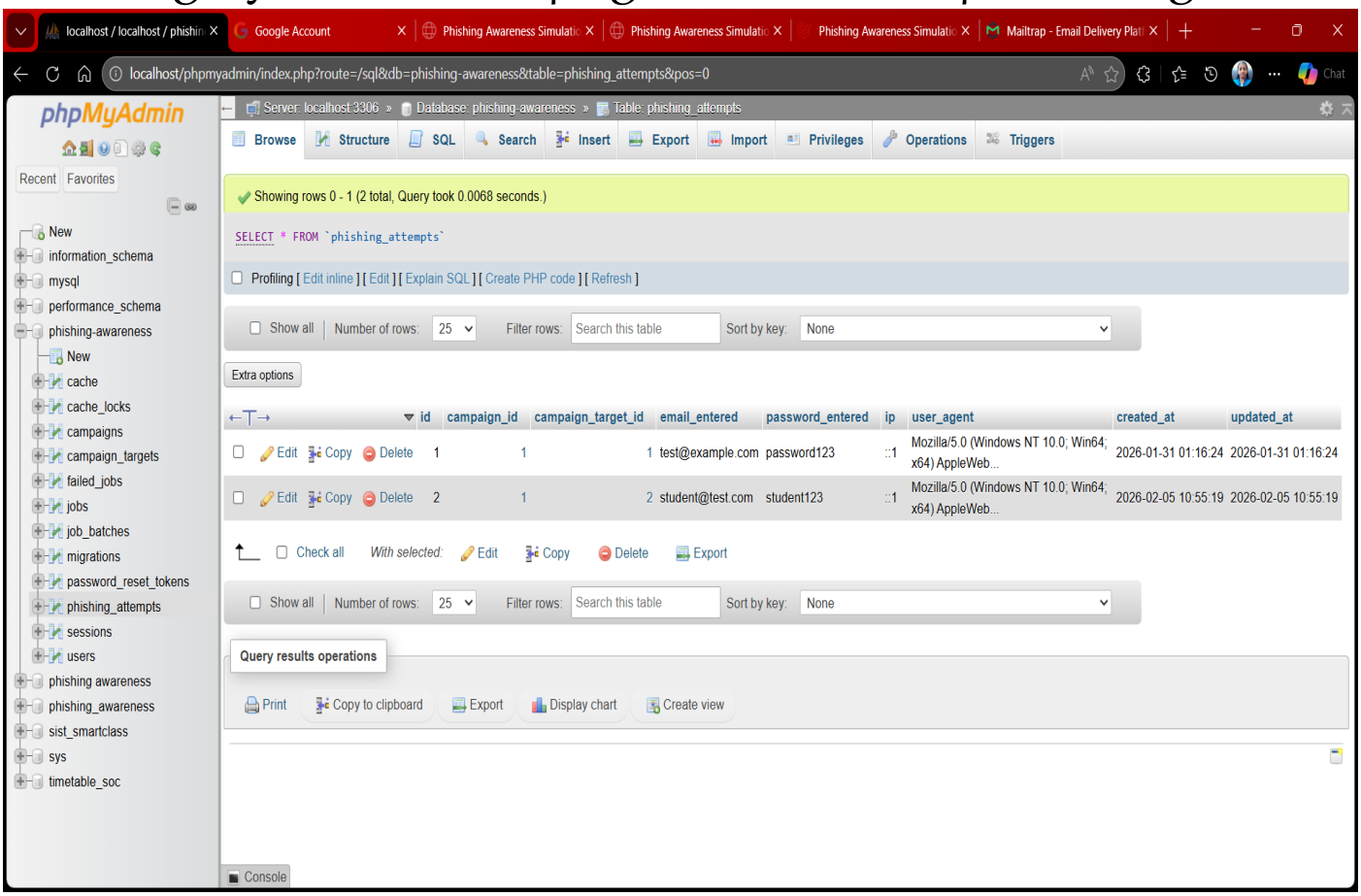


[Figure 1 - Laravel project running in local environment]

3. TASK 1: ENVIRONMENT & DATABASE SETUP

The project began with the initialization of the Laravel environment and the configuration of the relational database.

- **Database Creation:** A dedicated schema phishing-awareness was created.
- **Migrations:** Laravel migrations were executed to build the campaigns, targets, and phishing_attempts tables.
- **Verification:** Table structures were verified to ensure relational integrity between campaigns and their respective targets.

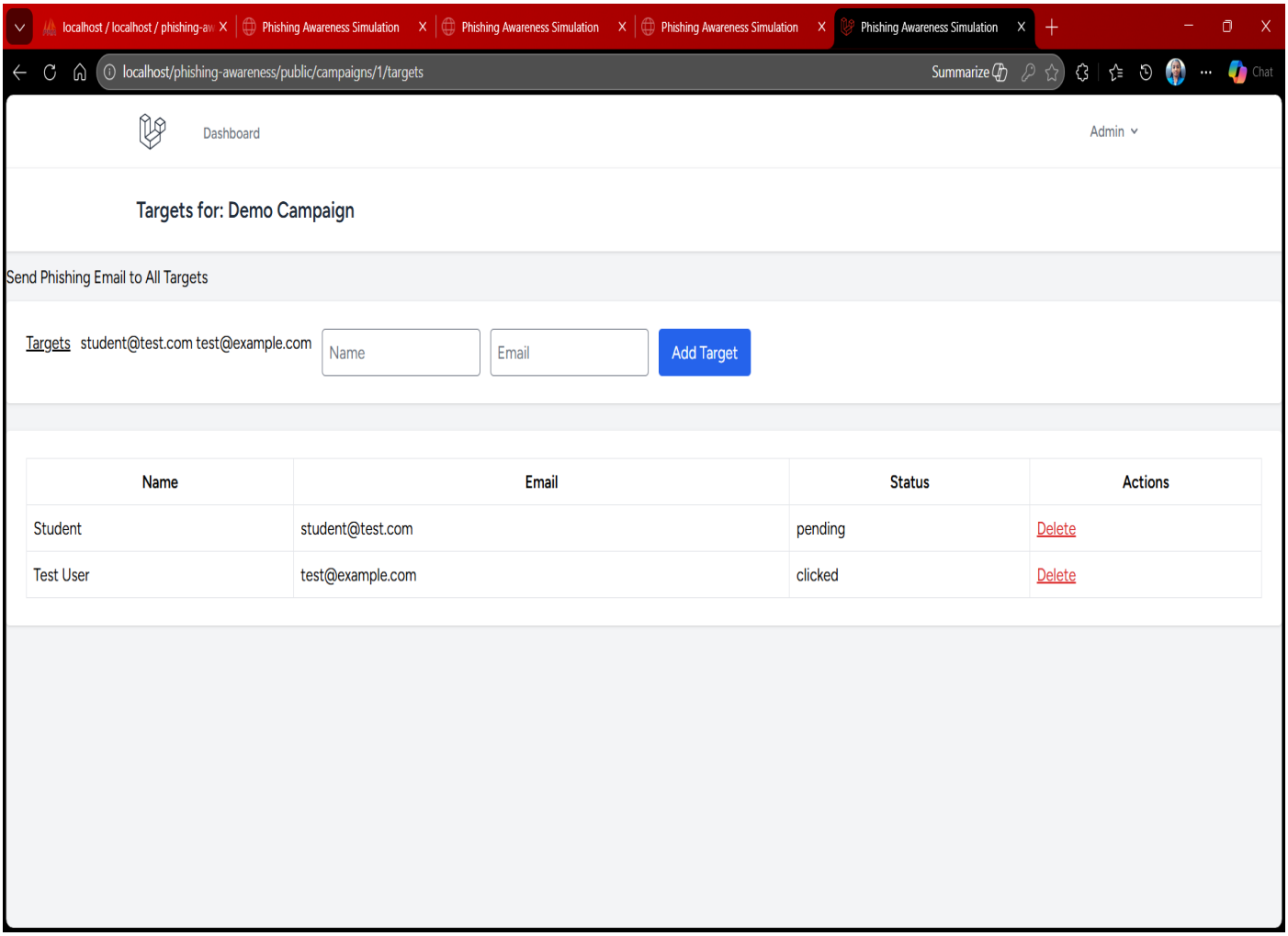


[Figure 2 - Database tables verified in phpMyAdmin]

4. TASK 2: CAMPAIGN MANAGEMENT (CRUD)

A central administrative dashboard was implemented to manage the lifecycle of a phishing simulation.

- **Functionality:** The system allows administrators to **Create, Read, Update, and Delete** campaigns.
- **Targeting:** Each campaign tracks specific targets through unique, tokenized URLs to ensure individual click-tracking accuracy.



[Figure 3 - Campaign and target management interface]

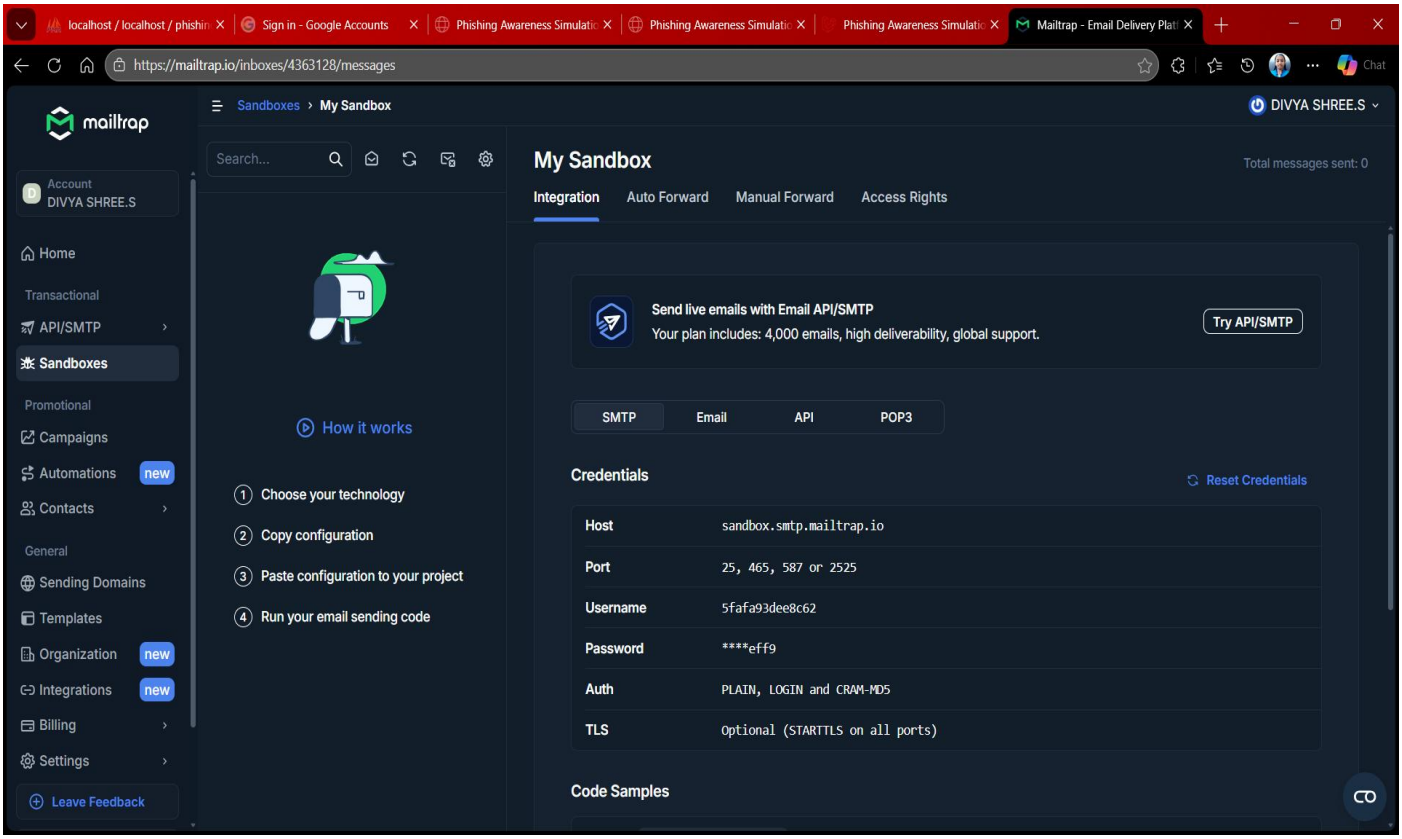
5. TASK 3: SMTP CONFIGURATION & EMAIL DELIVERY

To maintain ethical boundaries, the platform uses **Mailtrap** as an SMTP relay.

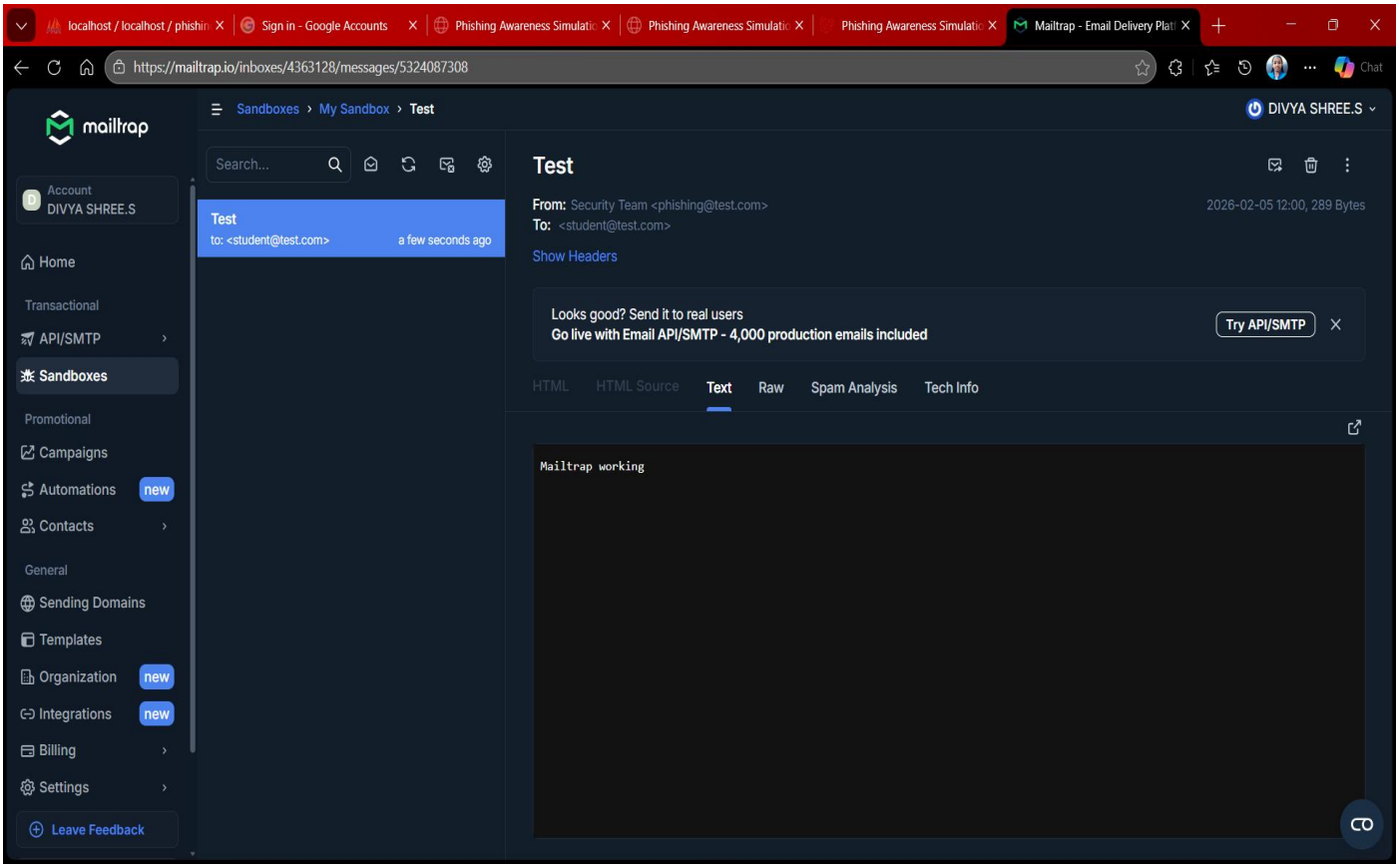
Configuration: The .env file was updated with Mailtrap credentials.

Verification: The Laravel mail cache was cleared to ensure seamless integration.

Outcome: Phishing emails were successfully triggered from the dashboard and intercepted in the Mailtrap sandbox.



[Figure 4 - Mailtrap SMTP configuration details]



[Figure 5 - Phishing email received in Mailtrap inbox]

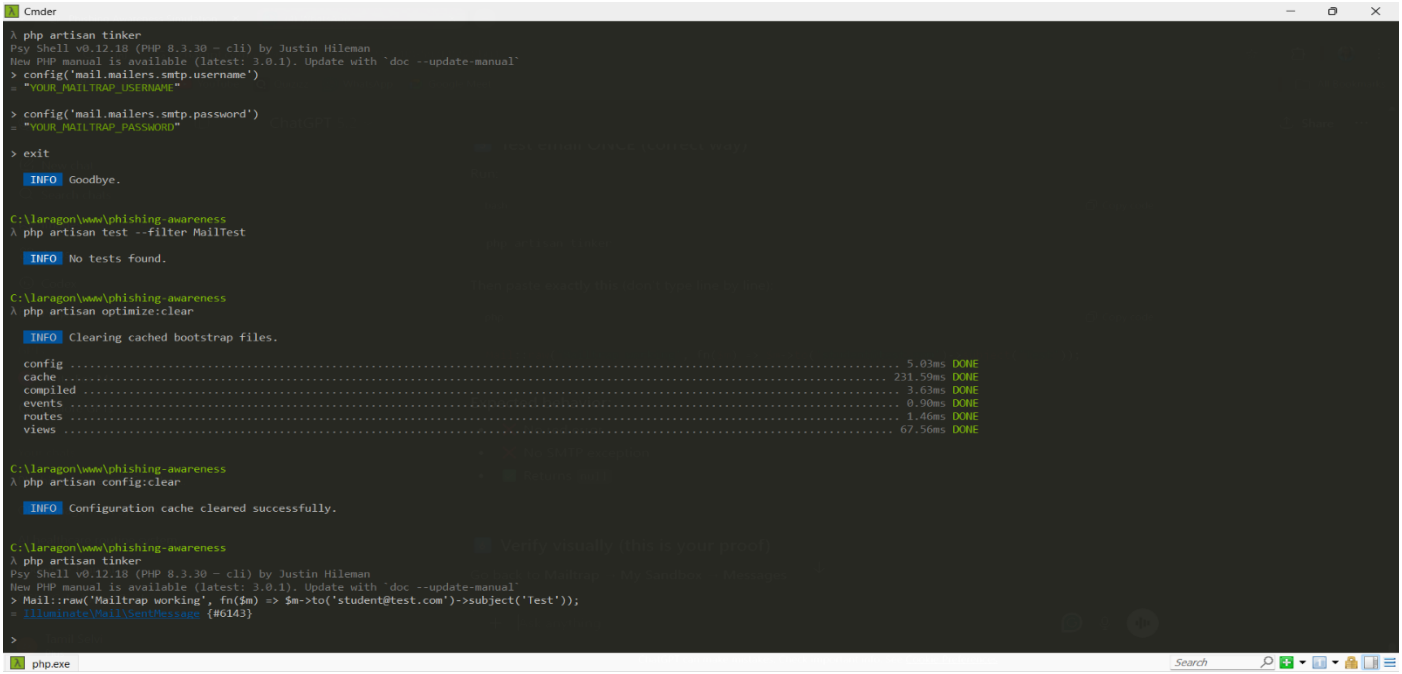
6. TASK 4: SIMULATED PHISHING & CREDENTIAL CAPTURE

The core "attack" logic mimics a real-world credential harvester.

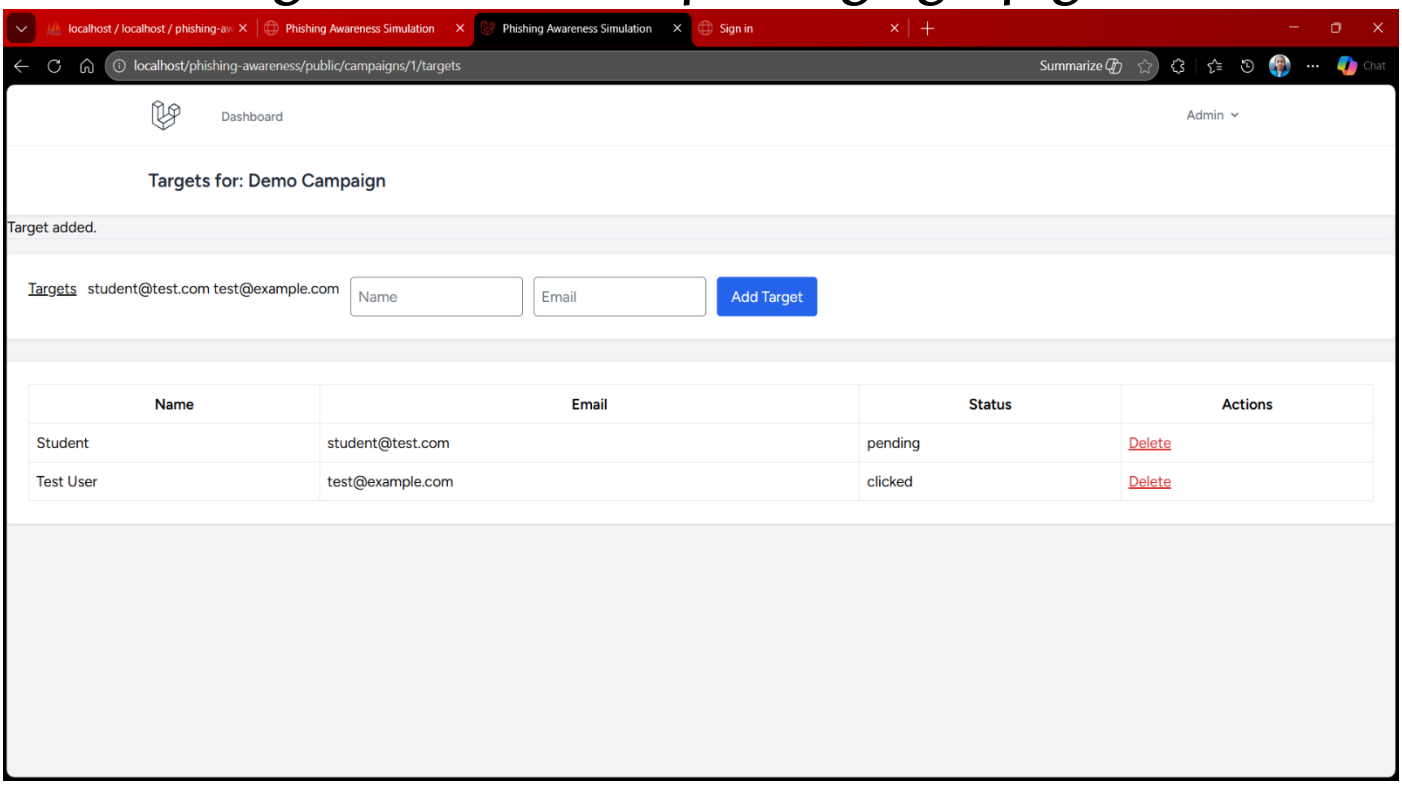
Landing Page: A high-fidelity simulated login page was designed to trick the target.

Interception: Upon form submission, the system captures the email and password (test data only) along with the target's IP address.

Redirection: To complete the simulation, the user is immediately redirected to a legitimate site (e.g., Google/Microsoft) to mask the capture process.



[Figure 6 - Simulated phishing login page UI]



[Figure 7 - Captured phishing attempt records in database]

7. RESULTS & CONCLUSION

The simulation successfully met all technical requirements:

- **Email Delivery:** 100% success rate in Mailtrap.
- **Click Tracking:** Verified via automatic "Clicked" status updates in the dashboard.
- **Data Integrity:** All attempts were logged with precise timestamps and metadata.

Conclusion: This project successfully demonstrates an end-to-end phishing awareness simulation using secure, sandboxed email delivery. All core objectives—email dispatch, user interaction tracking, credential capture, and database logging—were implemented and validated with verifiable evidence. The system accurately reflects real-world phishing attack behavior while maintaining strict ethical testing boundaries.

The project further proves the effectiveness of the Laravel framework in developing practical cybersecurity tools. The simulation provides a brutal but controlled insight into how easily users can be compromised, emphasizing the critical need for continuous cybersecurity awareness and training.